UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS

Faculty of Systems Engineering and Informatics



DOCTORAL PROGRAM IN DEEP TECH, ARTIFICIAL INTELLIGENCE & EMERGING TECHNOLOGIES

Research Methods and Scientific Integrity in AI and Advanced Technologies

DELIVERABLE No. 6

REPRODUCIBILITY AUDIT REPORT OF A PUBLISHED AAC STUDY

Audit scope: evidence and ratings are based on the published article and the Session 6 reproducibility scorecard; no independent re-analysis of participant-level data or source code is claimed.

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2026 – II

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REPRODUCIBILITY AUDIT REPORT OF A PUBLISHED AAC STUDY

This audit applies the Session 6 reproducibility scorecard to a 2026 context-aware augmentative and alternative communication (AAC) study. Because the audited paper is a single-case experimental study using a Wizard-of-Oz prototype rather than a trained stochastic machine-learning model, seed, split, and tuning criteria are interpreted according to their underlying reproducibility principle. The rating remains intentionally conservative whenever the paper does not provide enough information for an independent researcher to reconstruct the study.

EXECUTIVE SUMMARY

Griffen et al. (2026) compared a context-aware AAC prototype with a traditional AAC interface in three autistic preschool children using an ABAB reversal single-case experimental design. The study is unusually transparent about its participants, phase structure, procedures, fidelity monitoring, interrater reliability, and effect-size calculation. It also explicitly acknowledges several important threats to interpretation, including fixed condition order, a narrow age range, the absence of social-validity measures, variation in intervention fidelity, and substantial interface differences between the two AAC conditions.

The reproducibility picture is nevertheless incomplete. The context-aware function was not implemented as a functioning AI system; it was emulated by a human "wizard." Source code, a versioned prototype build, analysis scripts, machine-readable trial-level data, software versions, and an executable environment are not provided. Statistical reporting includes visual analysis and nonoverlap of all pairs (NAP) effect sizes, but no confidence intervals. External validation is absent. Using a transparent course-aligned heuristic (Yes = 2, Partial = 1, No = 0), the paper receives 10/18 points (55.6%). This number is descriptive rather than psychometric: the course scorecard was designed primarily for ML studies, so the written justification is more important than the percentage.

Overall judgment: the paper has moderate procedural transparency but limited computational reproducibility and limited external reproducibility. Its strongest value for future doctoral work is methodological: it shows why an actual functioning system, a

feature-matched comparator, a preregistered evaluation protocol, machine-readable data, open code, and independent validation are essential if claims are to be attributed specifically to context-aware intelligence rather than to interface simplification or human facilitation.

1. AUDITED PAPER AND SCOPE

1.1 Paper metadata

Field	Audited paper
Title	An Examination of Context-Aware Augmentative and Alternative Communication with Automated Response Options for Academic Communication From Autistic Children
Authors	Brenna Griffen; Sheida K. Raley; Christine Holyfield; Stephen MacNeil; Elizabeth R. Lorah; Jenna Burns
Journal	Journal of Special Education Technology
Publication	2026, 41(3), 407–419
DOI	https://doi.org/10.1177/01626434261420393
Study design	ABAB reversal single-case experimental design
Participants	Three autistic young children; AAC users / emerging symbolic communicators
Comparator	Traditional LAMP AAC color page vs. context-aware prototype
Technology status	Wizard-of-Oz prototype; context-aware behavior was human-controlled rather than generated by a functioning AI model

Field	Audited paper
Primary outcome	Accurate color-label response, coded together with response-time category
Primary analysis	Visual analysis of single-case data plus NAP effect size
Ethics	University IRB approval and guardian informed consent reported
Data availability	Article states that all data generated or analyzed are included in the published article

1.2 Audit scope and interpretation rule

The Session 6 scorecard asks about seeds, data splits, leakage, tuning separation, statistical tests, effect size and confidence intervals, external validation, computational documentation, and code/data availability. Three of those concepts—random seeds, train/validation/test splits, and hyperparameter tuning—do not map directly onto this paper because no stochastic ML model was trained. Rather than pretending that they do, this audit evaluates the closest methodological analogue and labels the limitation explicitly. This preserves the intent of the scorecard without penalizing the article for not being a conventional ML benchmark paper.

2. EXPERIMENTAL SETUP

The study enrolled three children and used an ABAB reversal design. Condition A used the LAMP AAC application opened to its color page. Condition B used a web-based context-aware prototype on the same large iPad Pro. In Condition B, four response options appeared in near real time as the communication partner stated them. Crucially, the context-aware behavior was implemented through a Wizard-of-Oz procedure: a researcher manually cued the options so that the interface appeared to perform automated speech recognition and response generation.

Each participant completed two phases in each condition. Phase transitions occurred after a preset five trials, for a total of 20 trials per participant across 3–4 sessions during

approximately two weeks. The interventionist asked color-identification questions during art activities, and the participant had up to 30 seconds to answer. The study recorded response accuracy and response-time category. Procedural fidelity was collected in all sessions and averaged 92.6% across blocks. A naïve independent coder reviewed all study videos; agreement was 100% for accurate versus inaccurate responding and 85.9% for response-time coding.

Analysis followed single-case experimental design conventions. Graphs were visually assessed for level, trend, slope, variability, and immediacy of effect. NAP was used as an effect-size measure. Across all participants, accurate responding was 20% under traditional AAC and 80% under context-aware AAC; the collective NAP was 0.827, interpreted by the authors as a moderate-to-large effect.

3. REPRODUCIBILITY SCORECARD

Scoring rule used for the requested overall score: Yes = 2 points, Partial = 1 point, No = 0 points. The course material does not prescribe a numerical conversion, so this is an explicit heuristic used only to summarize the item-level audit. The evidence and justification below remain the primary result.

Checklist item	Rating	Evidence from the paper	Audit interpretation
Seeds reported, many runs?	**PARTIAL**	No random seeds are relevant because no stochastic model was trained. Repeated ABAB phases and repeated trials provide repeated measurement, but not seed-to-seed robustness.	Some evidence is present, but the criterion is incomplete, only indirectly applicable, or insufficient for independent reproduction.
Splits	**YES**	The experimental partition is clearly	The paper reports

Checklist item	Rating	Evidence from the paper	Audit interpretation
described?		specified: A/B conditions, two phases per condition, five trials per phase, preset phase transitions, and 20 trials per participant.	enough information to satisfy the underlying reproducibility principle for this study design.
Leakage checked?	**PARTIAL**	Classic train/test leakage is not applicable. However, fixed condition order, prior AAC exposure, human Wizard control, and non-equivalent interfaces create analogous risks of contamination/confounding; some are acknowledged but not fully eliminated.	Some evidence is present, but the criterion is incomplete, only indirectly applicable, or insufficient for independent reproduction.
Tuning separated?	**YES**	No model tuning was conducted during evaluation. Phase changes were preset after five trials, so there is no evidence of outcome-driven early stopping or iterative tuning on evaluation results.	The paper reports enough information to satisfy the underlying reproducibility principle for this study design.
Statistical test named?	**PARTIAL**	The paper names visual single-case analysis standards and NAP effect size, but does not use an inferential statistical test or quantify sampling uncertainty.	Some evidence is present, but the criterion is incomplete, only indirectly

Checklist item	Rating	Evidence from the paper	Audit interpretation
			applicable, or insufficient for independent reproduction.
Effect size and CI?	**PARTIAL**	NAP effect sizes are reported for each participant and collectively; confidence intervals are not reported.	Some evidence is present, but the criterion is incomplete, only indirectly applicable, or insufficient for independent reproduction.
External validation?	**NO**	No independent site, later period, new population, or external replication is reported.	The criterion is not demonstrated in the paper.
Compute documented?	**PARTIAL**	The iPad, prototype concept, timing procedure, and Wizard mechanism are described, but software versions, web-app build, operating environment, source revision, and analysis runtime are not documented.	Some evidence is present, but the criterion is incomplete, only indirectly applicable, or insufficient for independent reproduction.
Code and data	**PARTIAL**	The article states that all	Some evidence is

Checklist item	Rating	Evidence from the paper	Audit interpretation
available?		generated/analyzed data are included in the publication, but no machine-readable dataset, code repository, prototype source, analysis script, or environment file is provided.	present, but the criterion is incomplete, only indirectly applicable, or insufficient for independent reproduction.

Score summary: 2 Yes × 2 points + 6 Partial × 1 point + 1 No × 0 points = 10/18 = 55.6%. The result should be read as "partially reproducible under the course scorecard," not as a validated quality index.

4. DETAILED AUDIT AND EVIDENCE

4.1 Seeds and repeated runs — PARTIAL

The study does not train a stochastic machine-learning model, so seed reporting is not directly applicable. The paper nevertheless contains repeated observations: two phases per condition and five trials per phase for each participant. This supports within-participant replication of the condition contrast, but it cannot answer the ML question "would the same algorithm produce the same result under a different seed?" A future fully automated context-aware system would need deterministic seeds or repeated-seed evaluation for any stochastic ranking, speech, or prediction component.

4.2 Experimental partitions — YES

The design is described with enough detail to reconstruct the phase structure. The paper specifies an ABAB reversal design, identifies Condition A and Condition B, states that every phase contained five trials, and states that phase shifts occurred after a preset five trials. This is a strong reproducibility feature because the stopping rule is explicit and not selected after observing the outcome.

4.3 Leakage and contamination — PARTIAL

There is no train/test pipeline, so classic ML leakage is not the correct label. The closest threats are design contamination and confounding. All participants began with the traditional condition, prior exposure to traditional AAC varied across participants, the same interventionist delivered sessions, and the context-aware interface differed from the comparator in the number, size, and style of icons. The authors explicitly acknowledge possible sequence effects and interface differences. Independent video recoding by a naïve coder reduces measurement bias, but the causal contribution of "context awareness" cannot be isolated from interface simplification.

4.4 Tuning separation — YES

No hyperparameter tuning or iterative model selection is reported because the prototype is not an operational AI model. Importantly, condition changes were fixed in advance after five trials rather than triggered by emerging performance. This reduces the risk of peeking-based stopping. For a future automated system, model development and participant evaluation should be separated explicitly with a locked version before the first outcome trial.

4.5 Statistical analysis — PARTIAL

The paper names the single-case visual-analysis framework and reports NAP, which is appropriate as an effect-size technique for single-case research. However, the analysis does not provide inferential uncertainty around the estimated effects. No confidence intervals accompany NAP, and no sensitivity analysis is reported. The absence of a conventional p-value is not itself a flaw in a single-case experimental design; the limitation is the lack of uncertainty quantification beyond visual patterns and point effect sizes.

4.6 Effect size and confidence intervals — PARTIAL

Effect-size reporting is a strength. The paper reports NAP values of 0.975 for Anna, 0.88 for Blake, 0.66 for Calvin, and 0.827 collectively. However, the scorecard asks for effect size together with a confidence interval. No confidence interval is reported for the NAP estimates, so the precision of the effect is not communicated.

4.7 External validation — NO

The study is intentionally preliminary. It includes three young participants, two local settings, a single academic task domain (color knowledge), and a short observation period. There is no replication with an independent research team, new site, later period, broader age range, or different functional communication contexts. The authors themselves call for broader and longer-term research.

4.8 Compute documentation — PARTIAL

The paper documents several implementation details: a large 12.9-inch iPad Pro, a web application, a Wizard-of-Oz controller, one-second cueing of options, and a phone stopwatch for timing. However, it does not identify the web-app version, source revision, operating system, browser, speech-recognition implementation because the AI was simulated, analysis software, package versions, or exact computational environment. These omissions prevent a computationally exact reconstruction.

4.9 Code and data availability — PARTIAL

The article includes the study data in figures and states that all data generated or analyzed are contained in the publication. This is better than an inaccessible data statement, but it is not equivalent to an open reproducibility package. There is no downloadable trial-level CSV, source code, analysis notebook/script, versioned prototype build, or environment file. A stranger could understand the experiment, but could not reproduce the software system from the article alone.

5. MISSING INFORMATION AND REPRODUCIBILITY RISKS

Missing / risk	Why it matters
Machine-readable trial data	The published graphs and narrative are informative, but an independent analyst should receive a trial-level dataset with participant, phase, condition, question, option order, response, accuracy, response-time category, fidelity item, and coder.

Missing / risk	Why it matters
Versioned source code and prototype build	The Wizard interface and participant-facing web application are not released with a commit hash or version identifier.
Executable analysis code	No script is supplied to reproduce the NAP calculations, figures, fidelity summaries, or interrater-reliability statistics.
Environment specification	Operating system, browser, application version, timing-device details, and analysis software versions are not reported.
Internal inconsistency in age reporting	Eligibility and the limitations section describe participants as four to five years old, while Table 1 lists Blake as 3 years 10 months. This discrepancy should be clarified because participant eligibility is a reproducibility-critical inclusion criterion.
Comparator equivalence	The context-aware display had four large cells and only the four spoken options, while the LAMP comparator displayed a broader color page. The authors acknowledge that icon number, size, and style differed. This prevents clean attribution of the observed effect to context awareness alone.
Condition-order control	All participants started in the traditional condition. Reversal helps, but counterbalancing or randomized starting condition across a larger sample would more directly reduce sequence effects.

Missing / risk	Why it matters
Independent implementation	The interventionist developed the procedures and the Wizard operated the experimental system. Although independent video recoding is a strength, a replication by an independent team would better establish procedural reproducibility.

6. OVERALL JUDGMENT

Overall reproducibility judgment: PARTIAL / MODERATE PROCEDURAL TRANSPARENCY, LOW COMPUTATIONAL REPRODUCIBILITY.

The study is strong at describing what human participants experienced. The ABAB design, phase rule, task, response coding, procedural-fidelity procedure, independent recoding, and NAP analysis are sufficiently explicit for another AAC researcher to understand and approximately replicate the behavioral protocol. The authors also disclose important limitations rather than overstating the findings.

What is not reproducible is the central technological claim at software level. The “context-aware” function was simulated by humans, and the publication does not provide a versioned executable system, code, analysis script, or machine-readable dataset. Consequently, an independent team cannot reproduce the claimed context-aware mechanism itself from the publication. In addition, the condition interfaces differ in ways that may explain part of the performance gap, and the study has no external validation. The paper therefore provides credible preliminary behavioral evidence for a design concept, not reproducible evidence that a functioning AI context-aware AAC system independently produces the observed effect.

7. THREE PRIORITY RECOMMENDATIONS

1. Release a complete reproducibility package

Publish machine-readable trial-level data, analysis code, NAP calculations, figure-generation scripts, the participant-facing prototype, the Wizard/control interface, software

versions, and an environment file in a versioned repository with a permanent release identifier. This would convert the paper from descriptively transparent to independently re-runnable.

2. Isolate the causal contribution of context awareness

Use a feature-matched comparator with the same number, size, style, and spatial arrangement of icons, changing only the contextual personalization mechanism. Counterbalance or randomize starting condition where feasible, preregister the phase rule and primary outcome, and retain blinded independent coding. This would reduce sequence effects and interface confounding.

3. Validate a functioning system externally

Replace the Wizard-of-Oz simulation with the actual speech/context pipeline, lock the version before evaluation, and replicate on new participants, sites, age groups, communication functions, and later time periods. Report effect sizes with confidence intervals and include social-validity or participant-choice measures. This would test both technical reproducibility and real-world generalizability.

8. IMPLICATIONS FOR MY DOCTORAL RESEARCH

This audit directly informs the design of my proposed Spanish-language multimodal AAC research. I should not claim a contribution merely because contextual suggestions outperform a static interface. The evaluation must isolate the intelligent component from interface simplification. Therefore, the intelligent and static conditions should be feature-matched, while the contextual-personalization mechanism is the controlled difference. The system evaluated in the doctoral study should be a functioning implementation rather than a Wizard-of-Oz simulation, with human-in-the-loop controls that allow the user or caregiver to accept, reject, edit, ignore, or disable suggestions.

The study also reinforces the need to predefine participant-level grouping, primary functional-communication outcomes, stopping rules, analysis procedures, and a reproducibility package before outcome evaluation. In addition to communication accuracy or speed, my protocol should prioritize independent functional acts, diversity of communication functions, prompting level, user agency, and social validity. The software

version, data schema, model/configuration files, seeds where stochastic components are used, environment, and analysis code should be preserved so that an independent researcher can reproduce both the computational pipeline and the reported outcomes.

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APPENDIX A. EVIDENCE MAP

Audit issue	Article location	Evidence summary
Design	Method – Design	ABAB reversal; two A and two B phases; preset 5-trial phase shifts.
Prototype status	Materials – Context-Aware AAC Prototype	Wizard-of-Oz; humans cued the listed color options to emulate automated context awareness.
Procedures	Procedures	3–4 sessions over two weeks; 20 total trials; 30-second response window.
Fidelity	Procedural Fidelity	Collected for 100% of sessions; mean 92.6%, range 83.3%–100%.
Independent coding	Interrater Reliability	Naïve coder reviewed all videos; 100% agreement for accuracy and 85.9% for response time.
Analysis	Data Analysis	Visual analysis plus NAP effect size.
Effect sizes	Results	NAP: Anna 0.975; Blake 0.88; Calvin 0.66; collective 0.827.
Main outcome	Results	20% accurate responding under traditional AAC versus 80% under context-aware

Audit issue	Article location	Evidence summary
		AAC collectively.
Sequence limitation	Limitations	All participants started in traditional AAC; authors acknowledge possible sequence effects.
Interface confound	Limitations	Number, size, and style of icons differed between conditions.
External validity	Limitations	Very narrow age range and academic communication only; future research recommended.
Data availability	Data Availability	All data generated or analyzed are stated to be included in the published article.
Age inconsistency	Participants + Table 1 + Limitations	Text says age 4–5; Table 1 lists Blake as 3;10; limitation states all were between 4 and 5.